

Highlights

- Hyperledger Fabric chaincode gates every document release on the download path
- Fail-closed design: downloads drop to zero in a Fabric outage, no ACL fallback
- On-path ledger access checks add no significant delay over a database baseline
- Tuned ordering sustains 193 TPS, above a 1,000-user legal workload estimate
- Encrypted IPFS storage keeps ledger commit latency independent of file size